SEIT

Department of Technology

Study Guide

IKT200- Introduction to Intelligent Systems

Course Objectives

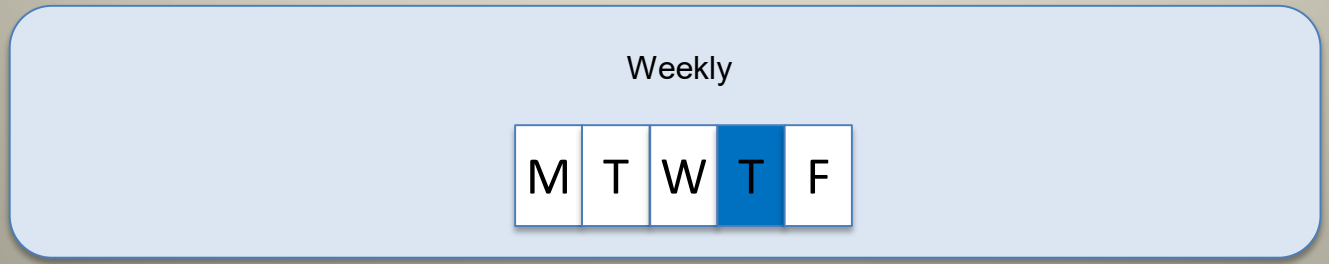
- Provide knowledge of business models, ecosystems and technologies used in intelligent systems.
- Enable students acquire knowledge about sensors, steering behaviour, networks, infrastructure and applications.
- Give skills in evaluation and selection of appropriate technologies/sensors.
- The students will discuss ethical considerations within the field.

Learning outcomes

- Knowledge- the student:
  - has knowledge about **intelligent systems**
  - has knowledge about **Internet of Things-** sensors, actuators and controllers
  - has knowledge about **artificial intelligence-** agents, finite state machines, expert systems and path finding algorithms such as **A***
  - has knowledge of important **rules and regulations** within the field
  - has knowledge of the process for **manufacturing solutions** within intelligent systems, including: identification of user groups, disclosure of requirements, design, development, testing, production setting and maintenance / further development
- Skills- the student:
  - can explain what **objects** can be included in the Internet of Things
  - is able to apply **artificial intelligence** based on **finite state machines**, **expert systems** and **search algorithms**
  - is able to carry out a systematic and documented **comparison of technologies and systems**
  - is able to understand how intelligent systems can be **linked to a business**
- Competence- the student:
  - can reflect on **ethical issues** related to intelligent systems

Course plan

- Start T**hursday, August 20th at 9.15**
  - Afterwards, every Monday from **09.15 to 13.00**
- Assignment hand-out: **Wednesday, December 2nd?**
- Assignment hand-in: **Friday, December 4th?**



Lectures and activities

Week	Topic	Activity	Readings
34	Introduction, Motivation- Intelligent systems Sensors, Actuators, Controllers	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapter 1
35	Agents, Finite State Machines, Design principles for connected devices	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapter 2
36	Rule-based ("expert") systems Turing and Eliza, Internet Principles, Beacon technology	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapter 3
37	Overview of prototyping, Pathfinding (part 0)	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapter 4
38	Prototyping Embedded Devices, Pathfinding (part 1)	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapter 5
39	Guest lecture by company: robotics	Live Zoom (TBA)- lecture and discussion	
40	Prototyping the Physical Design, Pathfinding (part 2)	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapter 6

Lectures and activities …

Week	Topic	Activity	Readings
41	Steering Behaviors, Prototyping Online Components	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapter 7
42	Techniques for Writing Embedded Code, Business models	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapters 8 & 9
43	Guest lecture with workshop in machine learning	Live Zoom (TBA)- lecture and discussion	
44	Moving to Manufacture, Ethics	Live Zoom- discussion, group activities; lecture, assignments	Textbook, Chapters 10 & 11
45	Exam assignment, course summary, wrap-up	Summary, questions, feedback	

Assessment

- Exam type- ***Individual written home assignment***
- Duration- ***48 hours***
- Grading scale- ***Norwegian grading system (A-F)***
- Weighting- ***100 % of the overall grade***

Readings

Books:

- Greengard, Samuel. *The internet of things*. MIT press, 2015.
- McEwen, A., & Cassimally, H. (2013). *Designing the internet of things*. John Wiley & Sons.
- Millington, Ian, and John Funge. *Artificial intelligence for games*. CRC Press, 2009.
- Russell, S., & Norvig, P. (2010). Artificial intelligence: a modern approach.

Articles:

- Belk, R. (2020). Ethical issues in service robotics and artificial intelligence. *The Service Industries Journal*, 1-17.
- Sikder, A. K., Acar, A., Aksu, H., Uluagac, A. S., Akkaya, K., & Conti, M. (2018, January). IoT-enabled smart lighting systems for smart cities. In *2018 IEEE 8th Annual Computing and Communication Workshop and Conference (CCWC)* (pp. 639-645). IEEE.

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